WORKED SOLUTION

STEP III 1992, Question 3

Pure • Curves and Polynomial Roots

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q3

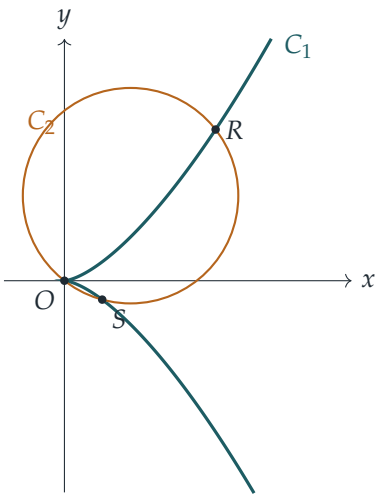
The key idea

Substitute the parametrisation straight into the general equation of a circle through the origin. This gives a polynomial equation in t whose roots are the parameters of the intersection points, and the origin shows up as a factor t^2. What is left is a quartic with no t^3 term, so its roots have a fixed sum and a fixed sum of products in pairs. Knowing two of the roots, r and s, then pins down the other two completely.

Solution

The sketch

Since $x = t^2 \geq 0$, the curve lies in the half-plane $x \geq 0$. Replacing t by $-t$ sends (x, y) to $(x, -y)$, so the curve is symmetric about the x-axis. Eliminating t gives $y^2 = x^3$, so $y = \pm x^{3/2}$. At $t = 0$ we have $\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}y}{\mathrm{d}t} = 0$, and near the origin $|y| = x^{3/2}$ is much smaller than x. So both branches leave O tangent to the x-axis, and the curve has a *cusp* at O. For large t the branches grow steeply, since $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3t}{2}$.



The curve C_1 is in teal. The amber circle is one example of C_2, the circle through O, R and S with $r = 1$ and $s = -\frac{1}{2}$.

The quartic

A circle through the origin has an equation with no constant term:

$$x^2 + y^2 + 2gx + 2fy = 0,$$

for real constants g and f, not both zero. The point with parameter t lies on the circle exactly when

$$t^4 + t^6 + 2gt^2 + 2ft^3 = 0, \qquad \text{that is,} \qquad t^2\left(t^4 + t^2 + 2ft + 2g\right) = 0.$$

The factor t^2 gives $t = 0$, which is the origin. Every intersection other than O has $t \neq 0$, so its parameter satisfies

$$t^4 + t^2 + at + b = 0, \qquad a = 2f, \quad b = 2g.$$

In particular r and s are roots of this equation, and a, b are real constants, as required.

Each point of C_1 has exactly one parameter, since $t = y/x$ when $x \neq 0$ (and $t = 0$ when $x = 0$). So distinct points R and S have distinct parameters, $r \neq s$, and the intersections of C_1 and C_2 correspond exactly to the real roots of the equation $t^2(t^4 + t^2 + at + b) = 0$.

The other two roots

Let the four roots of the quartic (real or complex) be r, s, p and q. Comparing coefficients of

$$(t - r)(t - s)(t - p)(t - q) = t^4 + 0 \cdot t^3 + 1 \cdot t^2 + at + b$$

gives

$$\begin{aligned} r + s + p + q &= 0, \\ rs + (r + s)(p + q) + pq &= 1. \end{aligned}$$

(Here we've grouped the six products in pairs as rs, pq and the four cross terms $rp + rq + sp + sq = (r+s)(p+q)$.)

Ask yourself We only know two of the four roots. How can we get at the other two without knowing a and b?

The two missing coefficients, t^3 and t^2, don't depend on the circle at all. The first equation gives $p + q = -(r+s)$. Substituting into the second,

$$pq = 1 - rs + (r+s)^2 = r^2 + rs + s^2 + 1.$$

So p and q are the roots of the quadratic

$$t^2 + (r+s)\,t + \left(r^2 + rs + s^2 + 1\right) = 0,$$

whose coefficients depend only on r and s. Any further intersection of C_1 and C_2 would have to have a parameter that is a real root of this quadratic.

(As a by-product, $a = -(r+s)(r^2+s^2+1)$ and $b = rspq = rs(r^2+rs+s^2+1)$, so the circle through O, R and S is determined by r and s.)

No further real intersections

The discriminant of the quadratic is

$$\begin{aligned}\Delta &= (r+s)^2 - 4\left(r^2 + rs + s^2 + 1\right)\\ &= -3r^2 - 2rs - 3s^2 - 4\\ &= -\left[2(r^2+s^2) + (r+s)^2 + 4\right].\end{aligned}$$

The bracket is at least 4, because each of $2(r^2+s^2)$ and $(r+s)^2$ is non-negative. So $\Delta \le -4 < 0$, and the quadratic has no real roots: p and q are a pair of non-real complex conjugates.

So the only real roots of $t^2(t^4 + t^2 + at + b) = 0$ are 0, r and s. Since each real root corresponds to exactly one point of C_1, the only real intersections of C_1 and C_2 are O, R and S. This is what we had to show.

Checks

Take $r = 1$, $s = -\frac{1}{2}$. The formulas give $a = -\frac{9}{8}$ and $b = -\frac{7}{8}$, so $f = -\frac{9}{16}$ and $g = -\frac{7}{16}$. The circle has centre $(\frac{7}{16}, \frac{9}{16})$, and both $R = (1,1)$ and $S = (\frac{1}{4}, -\frac{1}{8})$ are at squared distance $\frac{130}{256}$ from it, as is O. The leftover quadratic is $t^2 + \frac{1}{2}t + \frac{7}{4}$, with discriminant $\frac{1}{4} - 7 < 0$. This is the circle in the figure, and you can see it meets C_1 only at O, R and S. We also checked 400 random pairs (r,s) numerically. For each pair we found the circle through O, R, S, confirmed the factorisation, and scanned $-30 \le t \le 30$: the circle equation changed sign only at $t = r$ and $t = s$.

Common slips

- Forgetting that a circle through O has no constant term. The general circle has three constants, and losing the constant term is exactly what makes the factor t^2 appear.
- Thinking the double root $t = 0$ means the circle is tangent to C_1 at O. It comes from the cusp: every circle through O gives t^2 as a factor.
- Saying the discriminant is "clearly negative" without writing it as minus a sum of squares plus a positive constant.

Beyond STEP

The curve $y^2 = x^3$ is the standard example of a singular cubic curve. Counting intersections with multiplicity, as here, is the start of Bézout's theorem: curves of degrees m and n meet in mn points, counted properly. A cubic and a circle give 6, which matches our two at O, plus R, S and the two complex points. The same Vieta-type bookkeeping drives the group law on elliptic curves. Reid's *Undergraduate Algebraic Geometry* and Silverman and Tate's *Rational Points on Elliptic Curves* both take these ideas further.

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